From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

# FIITJEE JEE(Advanced)-2013

## ANSWERS, HINTS & SOLUTIONS
## FULL TEST –I
## (Paper-2)

| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | A | B | C |
| 2. | B | B | B |
| 3. | D | B | A |
| 4. | D | B | C |
| 5. | A, D | A, C | A, C |
| 6. | A, B, C, D | A, B, C | A, C |
| 7. | B, C | A, B, D | A, C |
| 8. | A, C | A, B, D | C, D |
| 9. | A, B, C | B, C | A, C, D |
| 1. | (A) → (p, q, r, s)<br>(B) → (q, r)<br>(C) → (q, r)<br>(D) → (p, q, r, s) | (A) → (r, s)<br>(B) → (p, r)<br>(C) → (r, t)<br>(D) → (q) | (A) → (s, t)<br>(B) → (p, q)<br>(C) → (r)<br>(D)→ (p, q, r, s, t) |
| 2. | (A) → (r)<br>(B) → (p)<br>(C) → (q)<br>(D) → (s) | (A) → (q, r, t)<br>(B) → (s, t)<br>(C) → (s, t)<br>(D) → (p, r, t) | (A) → (q)<br>(B) → (r)<br>(C) → (q)<br>(D)→ (p, q, r, s, t) |
| 1. | 8 | 5 | 4 |
| 2. | 4 | 2 | 8 |
| 3. | 9 | 2 | 3 |
| 4. | 5 | 5 | 5 |
| 5. | 3 | 8 | 3 |
| 6. | 1 | 8 | 5 |
| 7. | 5 | 6 | 3 |
| 8. | 8 | 7 | 3 |

# *Physics* PART – I

## SECTION – A

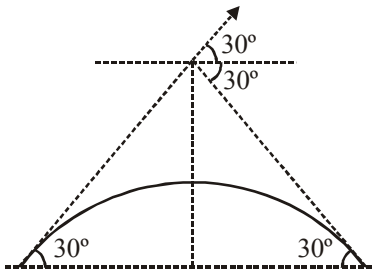
1. By conservation of momentum
$J = mV + 2mV$
$V = J/3m$.

2. $0 + 2 \times 1 + 3 \times 1 = V_B$
$\therefore\ V_B = 5$ volt

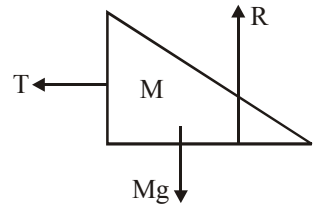
4. $S < \frac{1}{2}gt^2$
$\Rightarrow\ S < 2g$. (due to Lenz's Law).

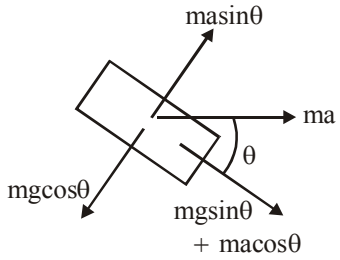
6. The molar heat capacity has the general definition
$C = \frac{1}{n} \cdot \frac{\Delta Q}{\Delta T}$
where n = number of moles, $\Delta Q$ = heat absorbed by the gas and $\Delta T$ = rise in temperature of gas. It is possible to obtain almost any set of values for $\Delta Q$ and $\Delta T$ by proper selection of a process.

7. $T = \frac{2u\sin\theta}{g} = \frac{2 \times 10 \times \frac{1}{2}}{10}$ = 1 sec.

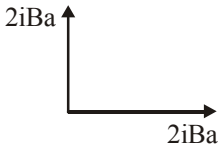



8. $M'g - T = M'a$ …(i)
$T = Ma$ …(ii)
$M'g = a(M + M')$
$a = \frac{M'g}{(M + M')}$
$ma\sin\theta = mg\cos\theta$
$a = g\cot\theta$
$g\cot\theta = \frac{M'g}{(M + M')}$
$\cot\theta M + \cot\theta M' = M'$
$M' = \frac{M\cot\theta}{(1 - \cot\theta)}$
T = M a
= M. g cot θ
$T = \frac{Mg}{\tan\theta}$.



9. $F_R = 2\sqrt{2}\,iBa$.




## SECTION –C

1. $P = \beta v^2$

$$\frac{mdv}{dt}.v = \beta v^2$$

$$\int_{v_0}^{2v_0} \frac{dv}{v} = \frac{\beta}{m}\int_0^t dt$$

$$\ln 2 = \frac{\beta}{m}.t$$

$$t = \frac{m\ln 2}{\beta}$$

$$t = \frac{4\times 2}{0.693}\times 0.693$$

t = 8 seconds.

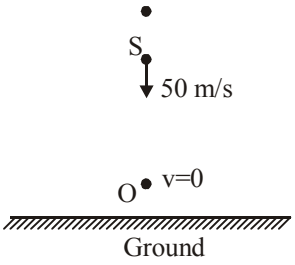
2. After 5 s, speed of detector = 50 – 10 × 5 = 0
and that of source = 0 + 10 × 5 = 50 m/s

and the source has fallen a distance $= \frac{1}{2}\times 10\times (5)^2 = 125$ m

and the detector has rises a distance

$= 50\times 5 - \frac{1}{2}\times 10(5)^2 = 125$ m

So, $f' = 130\left(\frac{300-0}{300-50}\right)$ = 156 Hz.




3. $g = \frac{GM}{R^2}$

$$g' = \frac{G.2m}{4R^2}$$

$$g' = \frac{g}{2}$$

$F = -mg'\sin\propto$

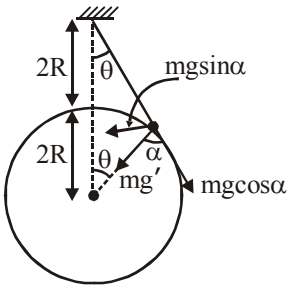
$$F = -\frac{mg}{2}.\sin 2\theta$$

As $\theta$ is much small

$$ma = -\frac{mg}{2}.2\theta$$

$a = -g.\theta$

$$a = -g.\frac{x}{2R}$$



$\therefore \quad \omega = \sqrt{\frac{g}{2R}}\, f$

$T = 2\pi\sqrt{\frac{2R}{g}}$

$T = 2 \times 3.14\sqrt{\frac{2 \times 6400 \times 10^3}{10}}$

$= 2 \times 3 \times 800 \times 1.41$

$T = 6768$ s.

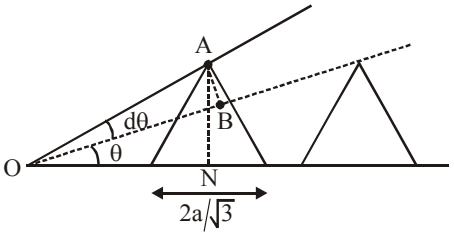
4. $AB = v\, dt \sin\theta$ …(i)

(along the circular arc)

$OA = \frac{AN}{\sin\theta}$

$ds = r\, d\theta$

given $vdt \sin\theta = OA\, d\theta$




$vdt\sin\theta = \frac{AN}{\sin\theta}.d\theta$

$vdt\sin\theta = \frac{a}{\sin\theta}d\theta$

$\frac{d\theta}{dt} = \frac{v}{a}\sin^2\theta$

$\frac{d\theta}{dt} = 5$ rad/s.

5. We have,

$10.2 = W + K_{max,\,1}$ …(i)

and $10.2\, Z^2 = W + K_{max,\,2}$ …(ii)

Also $\lambda_{de-Broglie} = \frac{h}{\rho} = \frac{h}{\sqrt{2mK}}$

$\therefore \quad \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{K_2}{K_1}} = 2.3 \quad \Rightarrow \quad K_2 = 5.25\, K_1$ …(iii)

Also $10.2\, Z^2$ = energy corresponding to longest wavelength of the Lyman series = $3 \times 13.6$

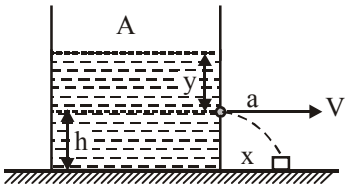
$\Rightarrow$ $Z = 2$.

$\therefore$ From equations (i), (ii) and (iii)

$W = 3$ eV.

6. Velocity of efflux $v = \sqrt{2gy}$

Range $x = \sqrt{2gy} \times \sqrt{\frac{2h}{g}}$




The velocity of the block must be $\left(\frac{dx}{dt}\right)$.

$\therefore \quad V_b = \frac{dx}{dt} = \sqrt{\frac{2h}{g}} \times \sqrt{2g} \times \frac{1}{2\sqrt{y}}\frac{dy}{dt}$

$V_b = \frac{\sqrt{h}}{\sqrt{y}}.\frac{dy}{dt}$ …(i)

Using equation of continuity

$$\frac{Ady}{dt} = a\sqrt{2gy} \qquad \ldots(ii)$$

equation (i) and (ii)

$$V_b = \sqrt{\frac{h}{y}} \times \frac{a}{A}\sqrt{2gy}$$

$$V_b = \sqrt{2gh} \times \frac{a}{A}$$

$$= 20 \times \frac{1}{20} = 1 \text{ ms}^{-1}.$$

7. If there were no hole

$$g_0 = \frac{GM}{R^2}$$

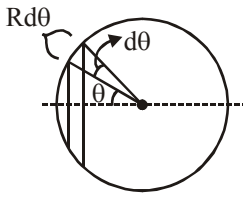
decrease in g due to hole (absent mass)

$$g_1 = \frac{GM/8}{\left(R/2\right)^2} = \frac{GM}{2R^2}$$

effective $g = g_0 - g_1$

$$= \frac{GM}{2R^2}$$

$= 5.5 \text{ m/s}^2$.

8. 


$$dq = \frac{q}{2\pi R}.Rd\theta = \frac{q}{2\pi}.d\theta$$

$$di = \frac{dq}{T} = \frac{qd\theta\omega}{2\pi 2\pi}$$

$$di = \frac{q\omega}{4\pi^2}.d\theta$$

$$dB = \frac{\mu_0 di(R\sin\theta)^2}{2R^3}$$

$$\int dB = \int_0^{\pi} \frac{\mu_0 \sin^2\theta}{2R}\left(\frac{q\omega}{4\pi^2}\right)d\theta$$

$$B = \frac{\mu_0 q\omega}{16\pi R}$$

$\phi = B\pi a^2$

$$\phi = \pi a^2.\frac{\mu_0 q\omega}{16\pi R}$$

$$\phi = \frac{\mu_0 q\omega a^2}{16R}$$

$$|\varepsilon| = \left|\frac{d\phi}{dt}\right|$$

$$|\varepsilon| = \frac{\mu_0 qa^2}{16R}\alpha \ .$$

$= 8$ volt

$i = \frac{8}{1} = 8$ A.

# Chemistry

**PART – II**

## SECTION – A

1. $W_{solute}$ = 3 gms, $W_{solvent}$ = 50 gm = 75 – 25

$$W_{OH^-} = \frac{3}{40} + \frac{x}{56}. \quad m = 5 = \frac{\left(3/40 + x/56\right)}{47} \times 1000$$

X = 9

2. Major product is formed via $E_2$.
H anti to halogen will be eliminated.

3. $n_{Hx} = n_{NaOH} = 0.25 \times V_{NaOH}$, $pH = pK_a = 5$ (acid buffer)

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C = 9 \qquad \Rightarrow \log C = -1 \Rightarrow C = 10^{-1}$$

$$C = \frac{0.25V}{60 + V_{NaOH}} \Rightarrow V_{NaOH} = 40 \text{ ml}$$

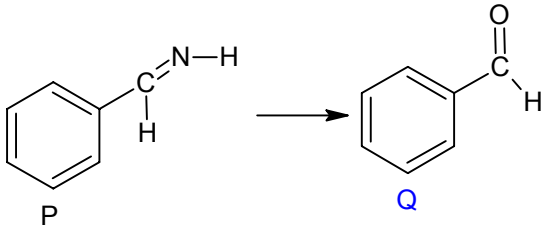
$n_{Hx}$ = 10 mm = 0.82 gm

$$\% = \frac{0.82}{1.025} \times 100 = 80\%$$

4. Time required for equifractional change in a first order reaction is constant.

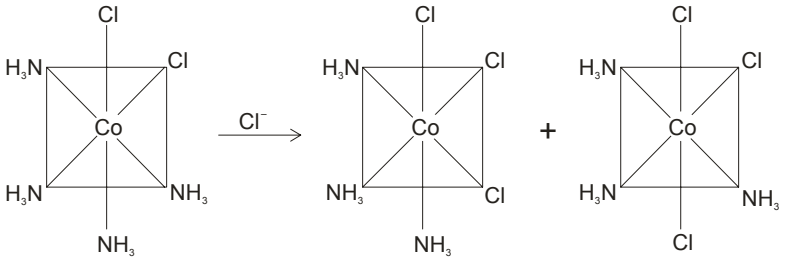
## SECTION – C

1.




P Q

2.




(Fac) (Mer)

3. $Na_2B_4O_7 + H_2O \rightarrow 2Na\left[B(OH)_4\right] + 2H_3BO_3$

1 mol 2 mol

$NaOH + H_5BO_3 \rightarrow \left[NaB(OH)_4\right]$

2 mol 2 mol

4. $A = 2y^{1/3}$ nm. $M = 6.023 \times y \times 4$ amu

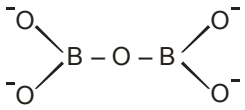
Density = $m/a^3 = \dfrac{6.023 \times 4 \times 1.66 \times 10^{-27}}{8y \times 10^{-27}}$ kg / m³

$= \dfrac{40}{8} = 5$ kg / m³

5. Two double bonds capable exhibiting geometrical and one chiral carbon molecule asymmetric.
$2^3 = 8$

6. $w = \dfrac{Pvm}{RT} = 58.33$ gm

$\% = \dfrac{58.33}{75} \times 100 = 77\%$

7. 

# Mathematics

## PART – III

### SECTION – A

1. PQ is a focal chord $Q\left(-\frac{1}{t}\right)=\left[\frac{a}{t^2}, -\frac{2a}{t}\right]$

Normal drawn at P(t) meets the curve at Q $\left[-t, \frac{-2}{t}\right]$

$\therefore$ Normal drawn at $Q\left(-\frac{1}{t}\right)$ meets the parabola again at R $\left[2t+\frac{1}{t}\right]$

$R\left[a\left(2t+\frac{1}{t}\right)^2, 2a\left(2t+\frac{1}{t}\right)\right]$

Equation of tangent at R is $y\left(2t+\frac{1}{t}\right)=x+a\left[2t+\frac{1}{t}\right]^2$ ..... (1)

Equation of focal chord PQ is $(t^2-1)y=2t(x-a)$ ..... (2)

T is the point of intersection of line (1) and (2)

$\Rightarrow T\left[\frac{a}{3}\left(4t^2-\frac{1}{t^2}\right), \frac{2a}{3}\left(4t+\frac{1}{t}\right)\right]$

$\therefore$ Area of $\Delta$ TQR = $\frac{8a^2}{3}\left[t+\frac{1}{t}\right]^3$

2. Let a point $(3\lambda + 1, \lambda + 2, 2\lambda + 3)$ of the first line also lies on the second line. Then

$\frac{3\lambda+1-3}{1}=\frac{\lambda+2-1}{2}=\frac{2\lambda+3-2}{3} \Rightarrow \lambda = 1$

Hence the point of intersection P of the two lines is (4, 3, 5). Equation of plane perpendicular to OP where OP is (0, 0, 0) and passing through P is $4x + 3y + 5z = 50$

3. For non triangle solution $\Delta=\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$

$\Rightarrow a + b + c = 0 \Rightarrow$ '1' is a root of $ax^2 + bx + c = 0$

4. Clearly we have

$\frac{xdy+ydx}{e^{xy}}=\frac{xdy-ydx}{x^2}$

$\Rightarrow \int d(xy)e^{-xy}=\int d\left(\frac{y}{x}\right) \Rightarrow \frac{y}{x}+e^{-xy}=c$

5. Let the roots of $az^3 + bz^2 + cz + d = 0$, $z_1 = x_1$, $z_2, z_3 = x_2 \pm iy_2$

$\Rightarrow z_1 + z_2 + z_3 = -\frac{b}{a}$

$\Rightarrow ab > 0$

Also $z_1 z_2 z_3 = x_1\left(x_2^2+y_2^2\right)=-\frac{d}{a}$

$\Rightarrow ad > 0$

6. On solving we have $\alpha = \frac{1}{2}$ and $\beta^2 = \frac{3}{4} \Rightarrow \beta = \pm\frac{\sqrt{3}}{2}$

$\therefore$ New roots are $\omega$ and $\omega^2$ which are roots of $x^3 - 1 = 0$ or $x^2 + x + 1 = 0$

7. Redefining the function, we get $f(x) = \begin{cases} \frac{x^2}{5} - \frac{4x}{5} + \frac{8}{5} & ; \quad x < 1 \\ 2 - x & ; \quad x \in [1, 2) \\ x - 2 & ; \quad x \geq 2 \end{cases}$

We have R.H.L at x = 1 is = $\lim\limits_{h\to 0} 2 - (1+h) = 1$

L.H.L at x = 1 is = $\lim\limits_{h\to 0} \frac{(1-h)^2}{5} - \frac{4(1-h)}{5} + \frac{8}{5} = 1$

$\therefore$ f(x) is continuous at x = 1

Now $f'(x) = \begin{cases} \frac{2x}{5} - \frac{4}{5} & ; \quad x < 1 \\ -1 & ; \quad 1 \leq x < 2 \\ 1 & ; \quad x \geq 2 \end{cases}$

R.H.D at $x = 1 \Rightarrow -1$

L.H.D at $x = 1 \Rightarrow -\frac{2}{5}$

$\therefore$ Not differentiable at x = 1

$\Rightarrow$ Not differentiable at x = 2

8. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ lie in the xy plane

Let $\vec{a} = \hat{i}$, $\vec{b} = -\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$ and $\vec{c} = -\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}$

$\therefore |\vec{p} + \vec{q} + \vec{r}| = |\lambda\vec{a} + \mu\vec{b} + \nu\vec{c}| = \left|\lambda\hat{i} + \mu\left(-\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right) + \nu\left[-\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right]\right|$

$= \sqrt{\left(\lambda - \frac{\mu}{2} - \frac{\nu}{2}\right)^2 + \frac{3}{4}(\mu - \nu)^2}$ $= \sqrt{\lambda^2 + \mu^2 + \nu^2 - \lambda\mu - \lambda\nu - \mu\nu}$

$= \frac{1}{\sqrt{2}}\sqrt{(\lambda-\mu)^2 + (\mu-\nu)^2 + (\nu-\lambda)^2} \geq \frac{1}{\sqrt{2}}\sqrt{1+1+4} = \sqrt{3}$

$\therefore |\vec{p} + \vec{q} + \vec{r}|$ can take a value equal to $\sqrt{3}$ and 2.

9. $z^2 + az + a^2 = 0$

$\Rightarrow z = a\omega, a\omega^2$ (where '$\omega$' is non real root of unity)

$\Rightarrow$ locus of z is a pair of straight lines and arg (z) = arg(a) + arg($\omega$) or arg(a) + arg($\omega^2$)

$\Rightarrow \arg(z) = \pm\frac{2\pi}{3}$ also, $|z| = |a||\omega|$ or $|a|\,|\omega^2| \Rightarrow |z| = |a|$.

## SECTION – B

1. Clearly lines are 2x + y + 2 = 0 and x + 2y + 1 = 0 and third line is bx + y + 5 = 0
For four circles three lines must not be concurrent $\Rightarrow b \neq 5$
For two circles exactly two lines should be parallel $\Rightarrow b = \frac{1}{2}$ and 2
For no circle three lines must be concurrent $\Rightarrow b = 5$
For infinite circles two lines must be identical $\Rightarrow b \in \phi$

2. (A) Solving $y = \frac{2K^2 - K}{3} > 0$ and $x = \frac{4K - 2K^2}{3} > 0 \Rightarrow K \in \left(\frac{1}{2}, 2\right)$

(B) Given $g(x) = \ln(\cos^{-1} x)$ ; As $0 \le \cos^{-1} x \le \pi \ \forall\ x$ [–1, 1]
So domain of g(x) = [–1, 1]. Hence number of integers are two (i.e. –1 and 0)

(C) Clearly, domain of expression = {–1, 1}. As x > 0 given
So x = 1
Hence value of expression $(1 + \sin^{-1} x)^{2010} = 1$

(D) Here $f(x) = \frac{x + a}{x + b}$ ….. (1)

and $f^{-1}(x) = \frac{a - bx}{x - 1}$ ….. (2)

Given $f(x) = f^{-1}(x) \Rightarrow \frac{x + a}{x + b} = \frac{a - bx}{x - 1}$

$\Rightarrow (1 + b)x^2 + (b^2 - 1)x - x(1 + b) = 0 \ \forall\ x \in D_f$
Hence b = –1 and $a \in R$

## SECTION – C

1. Using AM – GM inequality

$(a + b)^2 + (a + b + 4c)^2 = (a + b)^2 + (a + 2c + b + 2c)^2 \ge \left(2\sqrt{ab}\right)^2 + \left(2\sqrt{2ac} + 2\sqrt{2bc}\right)^2$

$= 4ab + 8ac + 8bc + 16c\sqrt{ab}$

$\therefore \frac{(a+b)^2 + (a+b+4c)^2}{abc}(a + b + c) \ge \frac{4ab + 8ac + 8bc + 16c\sqrt{ab}}{abc} \cdot (a + b + c)$

$= \left[\frac{4}{c} + \frac{8}{b} + \frac{8}{a} + \frac{16}{\sqrt{ab}}\right](a + b + c)\left[\frac{a}{2} + \frac{a}{2} + \frac{b}{2} + \frac{b}{2} + c\right] \ge 8 \times \left[5\sqrt[5]{\frac{1}{2a^2b^2c}}\right]\left(5\sqrt[5]{\frac{a^2b^2c}{2^4}}\right] = 100$

2. Putting X = x + 1 and Z = z + 1
$\therefore f(X) + f(Z) = f(X + Z)$

$\therefore f'(X) = \lim_{h \to 0} \frac{f(X + h) - f(X)}{h} = \lim_{h \to \infty} \frac{f(X) + f(h) - f(X)}{h} = \lim_{h \to 0} \frac{f(h) - f(0)}{h}$

$= f'(0) = 4 \Rightarrow f'(X) = 4 \Rightarrow f(X) = 4X + c$
$\therefore f(x) = 4x \Rightarrow f(2) = 8$

3. Let $f(\theta) = \prod(\tan\theta - \tan\alpha_i) - \left(\sum(\tan\theta - \tan\alpha_i)\right)$

We have $f(\theta)_{\theta \to -\pi/2} \to -\infty$

Again $f(\alpha_1) > 0$, $f(\alpha_3) < 0$ and $f(\theta)_{\theta \to \pi/2} \to \infty$

$\Rightarrow$ There will be 3 such values of '$\theta$'

4. Let the equation be $K(x-\alpha)(x-\beta)=0$
$\therefore$ Sum of coefficient = $K(\alpha-1)(\beta-1)$ is prime
$\Rightarrow$ K is 1 and $\alpha-1$ is 1
$\Rightarrow$ One root is 2 and $\beta-1$ is 2
Hence roots are 2 and 3.

5. Let $t=x^3$ in $I_2$

We have $I_2=\frac{1}{3}\int_0^1\frac{1}{e^t(2-t)}dt=\frac{1}{3}\int_0^1\frac{1}{e^{1-t}(1+t)}dt$

$\Rightarrow I_2=\frac{1}{3e}\int_0^1\frac{e^t}{(1+t)}dt=\frac{1}{3e}I_1\Rightarrow\frac{I_1}{eI_2}=3$

6. Let 3, 4 and r be radii of the circles inscribed into the $\Delta$'s ACD, BCD and $\Delta$ABC respectively, we get $\frac{r}{3}=\frac{AB}{AC}=\frac{c}{b}\Rightarrow b=\frac{3c}{r}$

Similarly $\Delta$ABC and $\Delta$BCD are similar, we get $\frac{r}{4}=\frac{c}{a}\Rightarrow a=\frac{4c}{r}$

Now as $c^2=a^2+b^2$

$\Rightarrow c^2=\frac{9c^2}{r^2}+\frac{16c^2}{r^2}\Rightarrow r=5$

7. Putting x = 2, $\frac{1}{2}$ and – 1 successively

$f(2)+f\left(\frac{1}{2}\right)=3$ ….. (1)

$f\left(\frac{1}{2}\right)+f(-1)=\frac{3}{2}$ ….. (2)

and f(–1) + f(2) = 0 ..... (3)

Solving, we get $f(2)=\frac{3}{4}$

8. On solving we have $A=\cos^4\theta+1-\cos^2\theta+1=\left(\cos^2\theta-\frac{1}{2}\right)^2+\frac{7}{4}\Rightarrow A\in\left[\frac{7}{4},\,2\right]$